NOVEMBER 5, 2026

Overnight D-Flow / QGF Results

A saved-evidence report. Scientific work is complete as of 07:30 Arizona; the 08:00 deadline and final archive publication are not claimed here.

Main finding. Longer critic training and an appropriate guidance scale restored useful control on native PointMaze. Some D-Flow variants are competitive; others fail. The robot remains far from solved. Stochastic inference does not, by itself, correct an unseen critic error.

What the experiments actually ask

Task family	What one trial means
Native PointMaze	A simulated point navigates a maze using actual MuJoCo transitions, up to 1,000 physical decisions. Task 1 aims at (20,20).
Native robot simulation	A robot moves the yellow cube to its goal while keeping pink and blue at theirs. All three must be within 4 cm.
Custom navigation	A separate 80-decision kinematic MDP with walls, goals and a clock. It is not the native OGBench benchmark.
Twenty synthetic point tasks	Choose one 2D point in a target region. Analytic flow + fitted reward critic; no next state or IQL V.

Three different learned quantities

Symbol	Meaning
Q(s,a)	Scalar critic value of an action. Guidance uses learned target-minimum twin Q.
V(s)	IQL scalar state value used in critic training: target r + γ × mask × V(s').
v(s,x,t)	BC flow vector velocity at internal pseudo-time t. It transforms a Gaussian source into an action; it does not estimate return.

Offline tuples → BC flow + IQL Q/V → sampler → action → native next state
Data contain (s, a, r, s′, mask). Frozen inference receives no new reward labels. Source iterations, flow slices and environment decisions are three different counters.

Three trained model seeds often share the same 20 reset/policy starts. A reported n/60 is descriptive; it is not 60 independent trained models or 60 distinct starts.

Evidence: TASKS_AND_METHODS.txt; TASK_AND_TRAINING_PIPELINE.png; branch reports named on page 7.

NATIVE POINTMAZE TASK 1

All eight D-Flow variants

Same frozen 1k-update actor and 50k-update Q/V per model seed (11, 23, 37). Each of the 16 fixed-λ conditions owns its validation-selected κ, η and source-iteration count.

Source updates	Jacobian	Q endpoint	λ = 0	λ = 1
Deterministic	Full	Full	52/60	52/60
Deterministic	Identity	Full	52/60	53/60
Deterministic	Full	Euler	0/60	0/60
Deterministic	Identity	Euler	53/60	58/60
Stochastic	Full	Full	55/60	51/60
Stochastic	Identity	Full	51/60	50/60
Stochastic	Full	Euler	51/60	12/60
Stochastic	Identity	Euler	53/60	52/60

Reference	Successes
Canonical BC; uniform BC population controls	0/60
QGF fixed κ=.02 / .04	0/60 / 0/60
QGF fixed κ=20 / own tuned κ	53/60 / 53/60
Greedy learned-Q grid: 81 actions per decision	51/60

Four model/condition primary choices are canonical-BC fallbacks. Separate positive-guidance rows remain visible. Deterministic full-Jacobian/Euler still scores 0/60 in that positive-only analysis.

The 58/60 row is a descriptive highlight among many families. Stochastic full/full λ=0 scores 55/60 vs QGF 53/60, yet its mean return is 36.58 lower. Own tuning and unequal compute prevent an isolated noise-effect or unbiased winner claim.

Tuning and accounting

Main κ grid: 0, .02, .04, .06, .08, .1, .2, .3, .4, .6, .8, 1, 2, 3, 5, 8, 12, 20, 40. D-Flow η: .005, .01, .03, .1; source iterations: 1, 2, 4, 8, 16, 32, 64, 128, 256. Stochastic temperature T=1. λ=0 and λ=1 are fixed columns, not a test-selected λ.

QGF checks all 19 κ values in native validation. Other families use learned-Q prescreens before native validation. Settings are best declared candidates, not global optima; QGF selects upper-edge κ=40 for all models.

3,912 validation episodes; 2,040 actual tests; 45,828 prescreen settings. The 106 display rows contain 4 positive secondary rows and 4 exact aliases; 102 unique physical configurations remain. No nonfinite/out-of-bounds test actions. Simulator validation is an oracle tuning resource, and compute is unequal.

Evidence: pointmaze_full_samplers/primary_aggregate.csv; SELECTED_SETTINGS.txt; pointmaze_full_samplers/selected_hyperparameters.csv.

What guidance changes

Final actions always use the full BC endpoint F(z), including its action clip. The Euler endpoint can be only a proxy for the gradient.

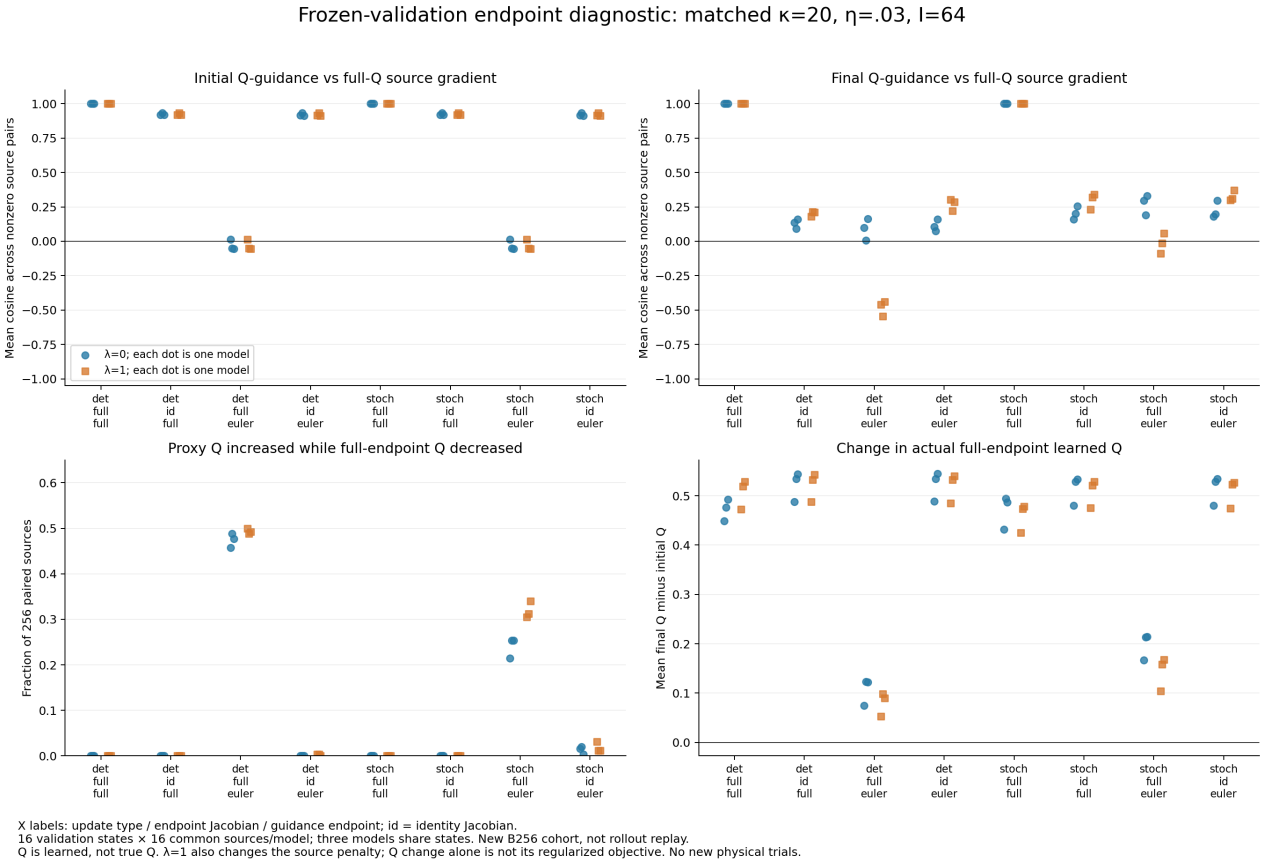
$$z_{i+1} = z_i + \eta[\kappa\, g_i - \lambda\, z_i] + \sqrt{(2\eta T)}\, \varepsilon_i$$

$$g_i = J_E(z_i)^T \nabla_a Q(s, E(z_i)), \text{ or the identity approximation } \nabla_a Q.$$

E is the full endpoint or an Euler estimate. Full-J differentiates through the endpoint clip; identity-J replaces the endpoint Jacobian by I. Noise is added only for stochastic updates.

κ scales Q guidance; **η** changes drift and noise scale; **λ** optionally pulls the source toward zero. λ is neither critic confidence nor noise strength. T=1 in the main stochastic comparison. A κ=0 shrinkage/noisy chain is not unchanged BC.

QGF recomputes each slice: velocity, current clean Euler endpoint and its Q gradient. The 8-slice update uses h=1/8. It does not reuse a stale first-step gradient.



Measured mismatch, not a causal failure proof. In 757 defined matched-setting pairs, 51.65% of full-J/Euler initial gradients oppose the actual full-endpoint Q gradient. A near-zero mean hides mixed directions. Proxy-Q-up/full-Q-down occurs in about 46-50% of deterministic pairs. No exact-zero-guidance/nonzero-full-gradient cases appeared in the 96-condition panel.

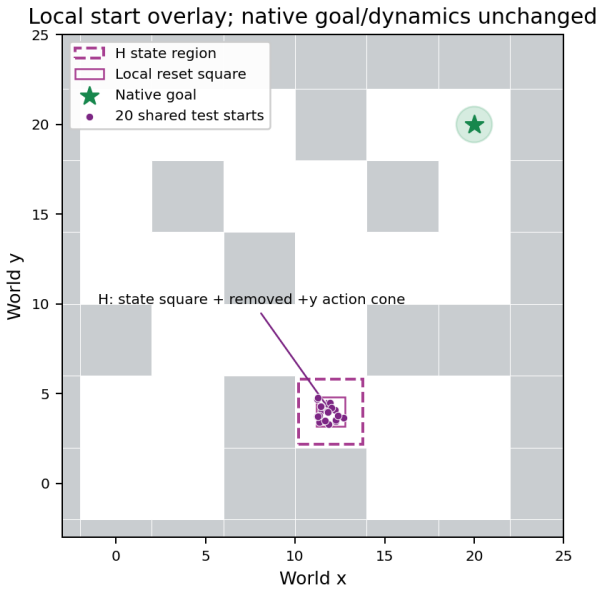
The panel reuses 768 unique saved validation state/source pairs across 24,576 endpoint comparisons. It adds no native physics or policy tests. λ=1 also changes the regularized objective κQ(F(z)) - ||z||²/2; Q alone is not that objective.

Evidence: endpoint_proxy_diagnostic/ENDPOINT_PROXY_REPORT.txt; all original-adapter action comparisons match bitwise.

Coverage must actually be visited

The intervention removes a joint state-and-action cone from Q/V training only. The BC actor stays trained on FULL data and may have seen those actions.

Default-start contrast missed H. All 660 tests avoided even its state region. Own tuned QGF scored FULL 54/60 vs HOLE 52/60; fixed κ=20 scored 54/60 vs 55/60. This did not directly test the intended missing-action region.



Local-H V2: a deliberate reset overlay

Starts: (12,4) ± .8 per axis; same native goal (20,20), dynamics, rewards and bounds. H: state-box radius 1.8 plus $a_y \geq .5$ and $|a_x| \leq .5$. Actions are never forced into the cone.

Every guided condition: 60/60
QGF and full/full D-Flow deterministic/stochastic × fixed λ=0/1, in BOTH FULL and HOLE.
Shared canonical BC: 0/60.

All 660 tests start inside H. Joint-cone execution: **593/600 guided + 60/60 BC** episodes. The exposure issue is addressed, but success is now at ceiling.

984 validation episodes; 660 actual tests; 33 physical cases; 30 primary aliases. All 30 choices globally freeze before tests: 24 D-Flow conditions tune κ/η/l; 6 QGF critic conditions tune κ only (η/l/λ N/A; 8 flow slices). D-Flow source iterations cap at 32. Both families use a four-band learned-Q shortlist. It is a changed-reset, posthoc stress variant, not a default benchmark or uniformly harder task.

FULL/HOLE compare independently retuned settings plus different critic data. The HOLE arm still retains 37,175 same-region actions outside the cone. Successful generalization is not proof of Q accuracy on every removed action or a stochastic advantage.

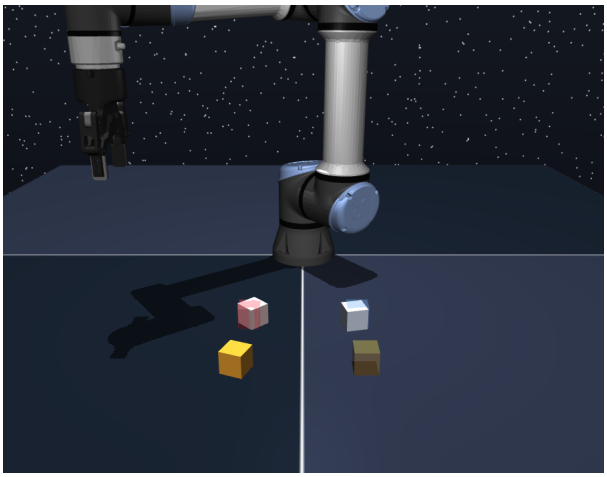
Failures are retained. V1 policy comparisons were not run after a timing-gate rejection. V2 is an explicit engineering-budget revision made before local outcomes. Checker-only arity/NumPy ULP failures were preserved; corrected audits and the predeclared B20 replay pass. No scientific models/settings changed after tests.

Evidence: pointmaze_critic_coverage_v1/COVERAGE_REPORT.txt; pointmaze_local_H_v2/REPORT.txt; pointmaze_local_H_v2/consumer_selected_settings.csv.

MUJOCO CUBE TRIPLE SIMULATION

Robot control is still weak

Move yellow to its translucent goal while keeping pink and blue at theirs. All three within 4 cm is native success; a cube may still be held.



Why the original floor was credible

The reduced 1M-transition pilot used small models and H1 control. A privileged native controller solved the same cohort 20/20. The slice contained no all-three-success states and only 557 joint configurations with the other two cubes at their goals.

These are geometric support counts, not full state/action density or proof that coverage caused every failure. Learned policies often failed before grasping.

Action chunks helped modestly, with limits

Model seed	H1: replan each step	H5: execute five controls
11	1/20	6/20
23	3/20	3/20
37	0/20	2/20
Descriptive total	4/60	11/60

Actors match starts and 60k updates, using the same original data plus 44 privileged TRAIN demonstrations (1,916 tuples). H5 has 31,257 parameters vs 26,117 for H1; capacity, temporal correlation and replanning frequency are not isolated. H5 has two collateral-damage episodes.

Better local Q calibration did not improve success. A proper 25D H5 critic and known-terminal action-invariance repair reduce uniform-action RMSE at four heldout already-goal states from 86.35 to .707 relative to matched IQL continuation. Yet all six physical settings still solve 2/20 (120 actual tests).

The original Q≈-205 at an already-successful pre-action state with exact target 0 is a genuine local Bellman residual. It does not label nonterminal Q or explain every grasp failure. Earlier actor repair gave 2/20 but worse mean return; adding pose-diverse demonstrations later gave 0/20 vs old 3/20.

Evidence: robot_horizon_control/HORIZON_STUDY_REPORT.txt; robot_chunk_critic/physical_critic_comparison/PHYSICAL_CRITIC_COMPARISON_REPORT.txt.

Correctors do not repair a wrong Q

All results below use frozen critics. More faithful sampling of their tilted target need not improve the real task.

Native task 1 method	Primary successes / 60
IPG / IPG-CV	51 / 45
Feynman-Kac corrector / residual-weighted IPG	47 / 52
MH proposals: full/full; identity/full; full/Euler; identity/Euler	51 / 46 / 51 / 51

What they change. IPG transports interacting particles; IPG-CV changes its control-variate estimate; IPG-RW retains residual weights; FK weights and resamples. Finite populations, ridge regularization, approximate flow scores and ancestry all matter. None provides a universal control gain.

The primary robot/maze particle implementation uses a declared piecewise-linear time extension between trained knots. It preserves original 8-slice velocities while avoiding unstable raw off-grid time derivatives. A continuous-flow CNF score differs from the finite clipped-Euler prior score; changing flow-step count can also change the prior map.

All four MH proposals fix λ=T=1 and accept against the same full-endpoint target:

$$\pi(z) \propto \exp[\kappa\, Q(s,F(z)) - ||z||^2/2]$$

Proxy/Jacobian choices change proposals and mixing, not this target. Finite chains need not mix; unadjusted finite-step Langevin is biased. No general normalizable source posterior is supplied by λ=0 noisy optimization.

Two additional native goals

Native goal	Tuned QGF	Fixed κ=20	BC / κ=.02,.04
Task 2	60/60	60/60	0/60
Task 5	49/60	47/60	0/60

Six fresh goal-specific critics; the same frozen actors. Actual 680 tests, 912 validation episodes and 16 planner controls. These are different native goals, not a uniformly harder-task series.

Other negative evidence: custom clock coverage improves local terminal calibration, but late-data control 58/60 trails matched early-data control 60/60 (BC 50/60). Toy MH full/full scores 181/400 vs ULA 151/400 and QGF 154/400 chosen points. Points and physical episodes are not pooled. An earlier broad IPG search remains incomplete after 523 completed validation settings and partial tests.

Evidence: pointmaze_task_variation/TASK_VARIATION_REPORT.txt; sampler_audit/SAMPLER_AUDIT.md; dflow_mh_sweep/DFLOW_MH_REPORT.txt.

What the evidence supports

This report describes a 07:30 scientific snapshot. It does not certify an 08:00 cutoff or say that a final archive already exists.

Actual units retained in the record

Completed branch	Validation	Actual heldout tests
Main native PointMaze comparison	3,912 episodes	2,040 episodes
Default critic-coverage contrast	888 episodes	660 episodes
Local-H V2	984 episodes	660 episodes
Native task 2 / task 5	912 + 16 planner checks	680 episodes
Matched robot H1/H5	48 episodes	120 episodes
Robot H5 physical critic comparison	64 new + 40 prior references	120 episodes
Custom clock coverage	1,032 episodes	420 episodes

These are branch-specific counts, not a pooled grand total. Paired starts, aliases, reused controls and exact replays must not be counted as independent new trials. The endpoint panel adds 24,576 comparisons on 768 shared pairs, not 24,576 episodes. The saved twin-head analysis reuses 4,080 probes across six critics and performs no new model queries.

What remains unresolved

Stochastic inference explores the frozen critic; it does not obtain new truth labels. Twin-head disagreement is not calibrated confidence: removing support makes gaps smaller in two of three saved-probe models. Higher learned Q, lower TD loss or better terminal calibration does not establish globally correct Q. Test-highlighted rows and unequal compute cannot establish an unbiased universal winner.

A useful next experiment - not executed. A predeclared 2×2 design could cross BC training coverage (FULL/HOLE) with Q/V training coverage (FULL/HOLE), using fresh cohorts and matched budgets. This would separate the current full-data actor support from critic-only coverage. It is a proposal, not a new result.

Reproduction status. Source/config/seed records, raw tuples/episodes, optimizer states, failures and audits are retained. The frozen-study inclusion and byte checks cover 4,775 payloads across 12 manifests as of this snapshot. Historical ZIPs remain immutable. No full relocated retraining or paper-scale reproduction was run.

The eventual snapshot excludes the runtime environment and recursive old ZIPs; copied Python dependency provenance is retained. Byte verification is portable. Some historic numeric entry points keep original-path and nanosecond-mtime guards. Use a working copy and the explicit verified timestamp restoration workflow. Fresh replication needs a new job, output and future deadline, not a completed historical resume.

Evidence: REPRODUCTION_GUIDE.txt; REPORT_INDEX.txt; final_verification/morning_claims_checklist.txt; frozen reports and manifests. All paths in this report are relative to outputs/overnight_research, except the robot photo in ../official_pilot/.